



## Best Interface is no Interface

*"The real problem with the interface is that it is an interface. Interfaces get in the way. I don't want to focus my energies on an interface. I want to focus on the job...I don't want to think of myself as using a computer, I want to think of myself as doing my job."*

— Donald Norman , 1990

Way back in 1999, BMW introduced a technology where a car's door unlocked when it sensed the key near it. There were sensors inside the key. This is just one simple application of invisible design. The concept of no UI is becoming increasingly popular. Recently, Roberts Mills released a book called "A guide to designing the invisible" that talks about methods that can be used to design devices with no visible interface. Golden Krishna, designer at Samsung America recently gave a talk on the topic of no UI after which the #NoUI hashtag became popular. Today, when designers are trying to answer every design problem by adding more intuitive interfaces and natural ways of interaction, all our devices are getting highly interface-driven. Almost every useful object – from washing machines and refrigerators to automobiles and even shoes and clothes come with interfaces on them. Most are operated by touch or gestures. Many user experience designers argue against this trend, though.

Consider this: you can choose between your current smartphone, which has an interface to put your phone on vibrate when you enter a meeting room, if you remember, or you can have a smartphone that automatically activates vibrate mode in meeting rooms, and actually goes to silent when the big boss is around. Which would you want?

However, this smartness has to be coupled with legible systems to understand needs better, or else you'll just have devices doing inexplicable things, like leaving you wondering, "When did I put my phone on silent?" The idea of no UI has

to be supported by emergency, alternate interfaces if needed in the event of panic. Consider an event from the movie "The Fifth Element". The character Korben gets up from his bed, his bed slides away automatically pulling off used linens to wash them and then pack them in plastic bags. However, now consider a situation where Korben goes to the restroom in the middle of the night, and then lands on his backside when trying to get back into bed. Thus a no UI-based systems need to be extremely context aware to perform well. Besides, you wouldn't want your bed to malfunction and try and wash and pack you along with your bed linen, would you? Thus we will need to have alternate emergency interfaces to signal a panic situation, where all such automatic actions are stopped immediately. Devices such as Nest, an intelligent thermostat that learns your patterns of temperature control, understand this need of alternate interfaces and provides you with an alternate interface to alter the temperature if needed manually. Designers believe that a device like Nest would be frustrating to use without this alternate emergency interface. What if your house is trying to cool itself on a hot day, but doesn't realise that you have a fever, and don't need the AC to work, because you want to sweat it out? The debate between UI and no UI settles on the point that having no UI leading to confusion and operational issues or an unusable beautiful looking UI are both bad choices. In the end, it depends on the type of application and the degree of control that can be taken away from the user. It is awesome to see such evolutionary developments in the space of UI design.

There's a lot we're looking forward to in the future, and only time will tell what the next big thing is. Until you get truly smart devices that just know what you want, even before you do, the best you can expect is to wave your hand and blink your eyes while expecting some device or another to interpret that as a command.



TELESAR V is a Robot that can be controlled with the human brain directly.

decides whether you want some or are just thinking randomly?

### Brain-to-brain

Won't it be awesome if we could communicate brain-to-brain without any interface in the middle. It would make things easy and crazy at the same time. Brain-to-brain communication was only a concept until recently, when a surprising research was published in Journal Scientific Reports. The neuroscientists could make two rats communicate using brain-to-brain communication devices, even when they were miles apart physically in 2 different cages. The rats could communicate and solve simple puzzles together. The encoded rat pushes a lever and earns a reward following which a brain implant records the cortex activity and delivers it to the decoder rat using neural links. The decoder then processes this information and performs a similar task. Now if you are still with us, this is just one proven way of how brain-to-brain interfaces have a credible future. The US BRAIN project is also working on thought helmets that soldiers can use to communicate amongst one another securely during war situations. Though brain-to-brain interfaces raise issues of super intelligence and other ethical concerns, scientists working in this field are sure that communicating over a neural link to replicate a task is very different from two brains planning a task together like a dual-core CPU. Just because we have super-computers doesn't mean we'll get super-brains by networking them, right? Err...hmm... 